

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) focused on GPU kernel development and ML systems. Wrote 6 CUDA kernels in C++ by hand for end-to-end MLP training on MNIST — manual cudaMalloc/cudaMemcpy, nvcc, ctypes FFI to Python — and extended gem5's C++ cache subsystem with a first-class LRU-IPV SimObject. Backed by a full RTL-to-GDSII ASIC flow on ASAP7 7nm and an open-source contribution merged upstream into Intel's reference architecture (**PR #655**) — kernel work grounded in how the silicon underneath actually behaves.

TECHNICAL SKILLS

GPU Kernels & CUDA: CUDA C/C++ (hand-written kernels) | nvcc | cudaMalloc/cudaMemcpy | Block/Thread Tuning | Python ↔ C++ FFI (ctypes) | PyTorch | TensorFlow | NumPy

Computer Architecture & Performance: gem5 (C++ SimObject internals) | Cache Hierarchy | 5-Stage Pipeline | SystemVerilog | Verilog | Cocotb | Synopsys Design Compiler | Cadence Innovus | Voltus | OpenROAD | Yosys | ASAP7 7nm | SkyWater 130nm

Languages, Tools & OSS: Python | C/C++ | Linux | Docker | Git | FastAPI | LangGraph | LangChain | Multi-provider LLM Routing | MCP Protocol | Intel reference architecture (PR #655 merged) | GitHub Actions CI/CD

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

May 2026

Tempe, AZ

Coursework: GPU Architecture (CUDA) · Advanced Computer Architecture · Algorithm/Hardware Co-Design · Artificial Intelligence · SoC Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

CUDA MLP — Hand-Written GPU Kernels for Neural Network Training

Fall 2024

- Implemented 6 CUDA kernels in C++ by hand for end-to-end MLP training on MNIST — **linear forward** (matmul + bias), **linear backward** (gradients w.r.t. inputs / weights / biases), and **ReLU forward/backward**; tuned block/thread launch geometry per kernel for the matmul and elementwise shapes, and managed host↔device data movement explicitly with cudaMalloc/cudaMemcpy rather than relying on a framework runtime.
- Compiled kernels via nvcc into shared libraries and bound them to Python through ctypes FFI for an end-to-end training pipeline that ran the full forward/backward/update loop from raw CUDA — not a framework wrapper — closing the loop from kernel code to a converged model on MNIST.

Intel Automated Self-Checkout — Open-Source Contribution (PR #655 Merged)

Jan – Feb 2025

- Contributed a sensor analytics microservice to **Intel's official self-checkout reference architecture** — configurable real-time data pipeline (MQTT, Kafka, HTTP) for LiDAR and weight-sensor streams with Grafana dashboard; **PR #655 merged upstream** after Intel engineering review — concrete OSS ownership across an external production-grade codebase.

LRU-IPV Cache Replacement Policy — gem5 C++ Internals (Performance Study)

Fall 2024

- Implemented LRU with Insertion Position Vector (IPV) replacement policy from scratch inside gem5's C++ source as a first-class SimObject, integrating all 4 cache interface methods (reset, touch, getVictim, invalidate); modified 5 gem5 source files and validated across 3 L2 cache configurations (128/256/512 kB), observing L2 replacements collapse from **637** → **112** → **2** as cache capacity grew — isolating capacity-vs-policy effects at 41.8M instructions, 2.04 IPC.

Graph Neural Network ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm

Spring 2025

- Designed an 11-module GNN ASIC accelerator in SystemVerilog (8 parallel MACs + 12-state FSM for COO edge aggregation) and drove the full sign-off flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT, Cadence Innovus place-and-route with CTS, Voltus post-APR power analysis — **50,300 μm^2 area, 52.29 mW power, 99.2 ns latency** — full intuition for what an AI accelerator looks like at the silicon level.

SiliconCrew — LangGraph Agent with Async Job Engine & Multi-Provider LLM Routing

Nov 2025 – Present

- Built a 21-tool LangGraph ReAct agent driving a complete RTL-to-GDSII flow; **async job engine** with ThreadPoolExecutor, per-run isolation, exponential backoff (30s → 10min), and on-disk recovery; multi-provider LLM routing (Gemini / OpenAI / Anthropic) with factory pattern + provider-specific quirk handling; full MCP server (stdio/SSE/HTTP); achieved **37/92 on CVDP** and **5/5 on ASU Spec2Tapeout**; public at github.com/naman-ranka/siliconcrew.

WORK EXPERIENCE

Systems Engineer

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won '**Best Use of AI**' at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.